Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21306

F-406

B.C.A. (Second Semester) Examination, 2024

(New Course)

Paper No. BCA-N-201

DIGITAL ELECTRONICS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Discuss various types of Logic Gates also discuss their applications.

   (b) Draw Circuit diagrams and truth table for following.

   (i) R-S flip flop

   (ii) D flip flop

   (iii) J-K flip flop

P.T.O.

2. (a) Using the K-Map method, simplify the following Boolean Function & obtain minimal SOP and minimal POS expression.

   $Y=\Sigma m(0,2,3,6,7)+\Sigma d(8,10,11,15)$

   (b) Discuss the difference between synchronous and asynchronous sequential circuit.

3. (a) What is the difference between multiplexer and demultiplexer? Explain in detail.

   (b) Explain excitation table, State table and state diagram.

4. (a) Show how a full adder can be converted to a full substractor with an addition of an inverter.

   (b) Explain the operations of master slave flip -flop and show how the race around condition is eliminated.

5. (a) Explain the various Boolean laws and simplify the following boolean expression.

(i) (A'+B+B'A)D

(ii) A'BC'+ABC'+A'B'C'+AB'C'

(iii) A'BC' +A'BC' D+BC'

(b) Give a detailed explaination using examples of conversion from binary to octal and octal to hexadecimal.

6. (a) Explain the difference between Encoder and Decoder. Draw circuit diagram for each.

(b) Define Multiplexer also draw circuit diagram for 8:1 multiplexer.

7. (a) Draw the internal diagram of a 4×1 multiplexer, clearly indicating the inputs and outputs. Explain the functionality using the function table.

(b) State and prove Associative law and distributive law with examples of each.

8. (a) Explain characteristic equation of JK flip flop from excitation table

(b) Design octal to binary encoder.

Total No. of Questions : 8] | Total No. of Printed Pages : 3

Paper Code : 21306

F-406

B.C.A. (Second Semester) Examination, 2023

(New Course)

Papar No. BCA-(N)-201

DIGITAL ELECTRONICS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. Perform the following conversions:
   (a) Decimal number 56 to its binary equivalent
   (b) Binary number 110110 to its decimal equivalent

P.T.O.

https://www.mjpruonline.com

   (c) Octal number 267 in its decimal equivalent
   (d) Hexadecimal number F3 to its binary equivalent
2. Discuss the concept of Karnaugh map. Solve the following Boolean function using Karnaugh map:
   F (A, B, C, D) =Σ (0, 1, 3, 5, 7, 9, 12, 13, 14)
3. Using De Morgan's laws, simplify the following Boolean expression:
   (A+B) + (C+D)
4. What is a binary to decimal decoder in digital electronics? Explain how it can be used to convert a binary number to its decimal equivalent.
5. What is the difference between a multiplexer and a demultiplexer in digital electronics? Provide examples of each.

https://www.mjpruonline.com

6. What is a JK flip-flop? Explain how it can be used to implement both synchronous and asynchronous operations.

7. Define the term "Race around condition" and explain how it can be avoided.

8. Explain the concept of "propagation time delay" in digital circuits. How does it affect the performance of a digital system?

https://www.mjpruonline.com
Whatsapp @ 9300930012
Send your old paper & get 10/-
अपने पुराने पेपर्स भेजे और 10 रुपये पायें,
Paytm or Google Pay से

https://www.mjpruonline.com

Paper Code : 21306

F-406

B.C.A. (Second Semester)

Examination, 2022

(New Course)

Paper No. BCA-N-201

DIGITAL ELECTRONICS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Fill in the blanks : 4

(i) $(1010.011)_2 = (\ldots\ldots\ldots)_{10}$

(ii) $(0.513)_{10}=(\ldots\ldots\ldots)_8$

https://www.mjpruonline.com

(iii) $(B65F)_{16}=(\ldots\ldots\ldots)_{10}$

(iv) $(153)_{10}=(\ldots\ldots\ldots)_8$

(b) What are universal gates? Why are they called universal? 3

(c) Explain with the help of block diagram the working principle of a full adder. 4

(d) Draw 2-input OR Gate using one 4×1 MUX. 3

2. (a) What is a flip-flop? What are the uses of flip flops? Explain it? 7

(b) Define computer organization, computer design and computer architecture. 7

3. (a) Explain the working of catch memory.3

(b) Explain the concept of virtual memory. 4

(c) Explain the various addressing modes used in computer system. 7

https://www.mjpruonline.com

4. (a) Design a 4-bit up-down counter. 7

(b) What is Register? Explain serial Register in detail. 7

5. (a) Differentiate b/w Combinational Digital circuits and sequential digital circuit. 7

(b) Distinguish b/w the following : 7

(i) RAM and ROM

(ii) Static and Dynamic memory

6. Write short notes on the following : 14

(a) CPU

(b) Demultiplex

(c) Adder

(d) Serial communication

7. (a) Show the logic diagram of clocked RS flip-flop with four NAND Gates. 7

https://www.mjpruonline.com

(b) Show the logic diagram of a clocked D flip-flop with AND and NOR Gates. 7

8. (a) What is the difference between serial and parallel transfer? What type of register is used in each case? 7

(b) Discuss various bus control methods. 7

https://www.mjpruonline.com
Whatsapp @ 9300930012
Send your old paper & get 10/-
अपने पुराने पेपर्स भेजे और 10 रुपये पायें,
Paytm or Google Pay से

https://www.mjpruonline.com

http://www.mjpruonline.com

Paper Code : 21306

F-406

B. C. A. (Second Semester) EXAMINATION, 2019

(New Course)

Paper No. BCA— (N)—201

DIGITAL ELECTRONICS

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) Perform the following conversions :

   (i) $(ABC)_{16} = (?)_{10}$

   (ii) $(47.6)_8 = (?)_{10}$

   (iii) $(5137)_{10} = (?)_{BCD}$

   (iv) $(62.7)_8 = (?)_{16}$

   (b) Find 9's and 10's complements of the following decimal numbers :

   (i) 2431

   (ii) 5299

http://www.mjpruonline.com

2. (a) What do you mean by the base of a number system ? Give examples to illustrate the role of base in positional number system.

   (b) What is logic gates ? Discuss the various types of gates. Why is the NAND gate called Universal gate ?

3. (a) Simplify the following Boolean function, using Karnough maps :

$$F(A, B, C, D) = \Sigma(0, 1, 2, 5, 8, 10, 13)$$

   (b) Express the Boolean function :

$$F = AB + AC + AD$$

   in sum of minters form.

4. (a) State and prove two basic De-Morgan's theorems.

   (b) Convert the given expression in standard POS form : http://www.mjpruonline.com

$$f(A, B, C) = (A + B)(B + C)(A + C).$$

   (c) What do you mean by general switching problem ?

5. (a) Implement a full adder and explain.

   (b) Design a full adder using NAND gates.

6. Differentiate between the following :

   (i) Multiplexer and De-multiplexer

   (ii) Combinational and Sequential circuit

   (iii) POS and SOP

   (iv) Analog and Digital signal

   (v) Serial and Parallel register

http://www.mjpruonline.com

http://www.mjpruonline.com

7. (a) Explain the working of master slave JK flip-flop.

   (b) What is race around condition ? Explain in brief.

   (c) How can one convert D flip-flop to T flip-flop ? Explain.

8. Write short notes on the following (any *four*) :

   (i) Floating point number representation

   (ii) Counters

   (iii) Venn diagram representation of Boolean algebra

   (iv) Signed and unsigned number representation

   (v) Encoders

   (vi) Weighted number systems

http://www.mjpruonline.com

http://www.mjpruonline.com

http://www.mjpruonline.com
Whatsapp @ 9300930012
Your old paper & get 10/-
पुराने पेपर्स भेजे और 10 रुपये पायें,
Paytm or Google Pay से

1700

http://www.mjpruonline.com

MJPRUonline.com

Paper Code : 21306

F–406

B. C. A. (Second Semester)
EXAMINATION, 2018

(New Course)

Paper No. BCA—N–201

DIGITAL ELECTRONICS

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) Write the truth table and symbol of following gates :
   (i) XOR
   (ii) NAND
   (iii) NOR
   (b) Represent decimal number 27 in binary form using :
   (i) BCD Code
   (ii) Excess-3 Code
   (iii) Gray Code

MJPRUonline.com

MJPRUonline.com

2. (a) Perform using 2's complement method :
   (i) 0011.1001 – 0001.1110
   (ii) 01100 – 00011
   (b) Minimize the four variable logic function using K-Map :

   $F(A, B, C, D) = \Sigma_m (0, 1, 2, 3, 5, 7, 8, 9, 11, 14)$

3. (a) Make a 32 : 1 MUX using 16 : 1 MUX.
   (b) Design a full adder circuit and realize it.
4. (a) What do you mean by Race around condition in JK Flip Flop ? How can it be solved ? Explain.
   (b) What is the purpose of counter ? Draw the logic diagram of 3-bit synchronous counter.
5. (a) State difference between Serial Registers and Parallel Registers.
   (b) Compare and contract state table and excitation table used in sequential circuits.
6. (a) Simplify the following expressions using Boolean Algebra :

   MJPRUonline.com

   (i) $AB + A(CD + \overline{CD'})$
   (ii) $\bar{A}'B + AB\bar{C}' + ABC$
   (b) Design a full subtractor circuit with three inputs *x*, *y* and *z* and two outputs D and B. The circuit subtracts the bits *x*-*y*-*z*, where *z* is the input borrow, B is the output borrow and D is the difference.

MJPRUonline.com

MJPRUonline.com

7. Design a combinational circuit with three inputs, x, y and z, and the three outputs, A, B, and C. When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7 the binary output is one less than the input.

8. Write short notes on any *four* of the following :

   (a) SR Flip Flop

   (b) Decoder and Demultiplexer

   (c) BCD Ripple counter

   (d) De Morgan's theorem

   (e) SOP and POS to design combinational circuit

   (f) Floating point representation

MJPRUonline.com

MJPRUonline.com